

COORDINATE-BLOCK FACTORIZATION UNDER SINGLE-SANDWICH DIVISION AND APPLICATIONS TO PROPERTY $(TT)/T$

GPT-5.6 SOL

ABSTRACT. We prove a coordinate-block factorization theorem for general linear groups over rings satisfying the single-sandwich condition

$$a \neq 0 \implies xay = 1 \text{ for some } x, y.$$

Every element of $GL_n(R)$ is a product of at most $2n + 2$ factors, each an elementary transvection or an element of one fixed coordinate copy of $GL_{n-1}(R)$. Thus $GL_n(R)$ has width at most $2n + 2$ relative to the elementary root subgroups and that fixed coordinate block.

The algebraic factorization is the main input developed here. We combine it with a quantitative root-subgroup estimate in Mimura's relative quasi-cocycle framework. The estimate is proved here by an exhaustive finite-word filtration whose boundary terms vanish by an exact telescoping identity. If R is a nontrivial finite-type $\mathbb{F}_2$ -algebra carrying a binary Leavitt family, satisfying the single-sandwich condition, and satisfying $\text{diag}(u, 1) \in E_2(R)$ for every $u \in R^\times$, then $E_n(R)$ has property $(TT)/T$ for every $n \geq 2$. For the binary Leavitt algebra this identifies every rank with the unit group and gives

$$E_n(L_{\mathbb{F}_2}(1, 2)) \cong L_{\mathbb{F}_2}(1, 2)^\times, \quad n \geq 2,$$

with property $(TT)/T$.

1. INTRODUCTION

Property $(TT)/T$, introduced by Mimura, asks that Hilbert-space quasi-cocycles be bounded after the invariant-vector part of the representation has been removed. It is stronger than Kazhdan's property (T) and is designed to control bounded cohomology and homomorphisms into groups admitting unbounded quasi-cocycles [7]. Ershov and Jaikin-Zapirain proved property (T) for elementary groups of rank at least three over arbitrary finitely generated associative rings [4]. The corresponding quasi-cocycle problem is subtler: the bounded error in the cocycle identity accumulates under a long word, so finite generation alone gives no global bound.

The present paper supplies a globalization mechanism for a class of purely infinite rings. Its algebraic input is an explicit matrix factorization. The

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hypothesis is the following strong two-sided division property:

$$(1) \quad a \neq 0 \implies \exists x, y \in R, \quad xay = 1.$$

We call (1) the single-sandwich property. Ara, Goodearl, and Pardo proved that, for a nondivision simple ring, this is equivalent to pure infiniteness [3, Theorem 1.6].

Theorem A (Coordinate-block factorization). *Let R be a nontrivial ring satisfying (1), let $n \geq 2$, and fix a coordinate $j < n$. Every $A \in \mathrm{GL}_n(R)$ admits a factorization of length at most $2n + 2$ in which every factor is either an elementary transvection or belongs to the coordinate copy*

$$\mathrm{GL}_{n-1}^{(j)}(R) = \{B \in \mathrm{GL}_n(R) : Be_j = e_j, e_j^t B = e_j^t\}.$$

Formal counterpart:

`PropertyTTPaper.coordinateBlock_factorization.`

Theorem A is a relative bounded-width statement rather than bounded elementary generation: the bounded factors come from the elementary root subgroups together with one fixed coordinate block. This is exactly the form required by the quasi-cocycle argument. By the characterization just cited, the conclusion applies in particular to every unital purely infinite simple ring.

For the synthesis theorem, a *binary Leavitt family* in R means elements x_0, x_1, y_0, y_1 satisfying

$$y_i x_j = \delta_{ij}, \quad x_0 y_0 + x_1 y_1 = 1.$$

It supplies the matrix self-similarity used to move between ranks. We say that an $\mathbb{F}_2$ -algebra is *finite type* when it is generated by a finite set as an $\mathbb{F}_2$ -algebra, equivalently when it is a quotient of $\mathbb{F}_2\langle X \rangle$ for some finite set X .

Theorem B (Rigidity synthesis). *Let R be a nontrivial finite-type $\mathbb{F}_2$ -algebra. Assume that*

- (1) R contains a binary Leavitt family;
- (2) R satisfies the single-sandwich property (1);
- (3) $\mathrm{diag}(u, 1) \in E_2(R)$ for every $u \in R^\times$.

Then $E_n(R)$ has property $(TT)/T$ for every $n \geq 2$.

Formal counterpart:

`PropertyTTPaper.finiteType_elementaryGroup_hasTTmodT.`

The assumptions play separate roles: finite type presents the ring as a quotient of a finite free algebra, the Leavitt family moves between matrix ranks, the single-sandwich property supplies bounded matrix reduction, and the diagonal condition identifies elementary and general linear groups.

Theorem C (Binary Leavitt algebra). *The unit group of the binary Leavitt algebra has property $(TT)/T$:*

$$L_{\mathbb{F}_2}(1, 2)^\times \quad \text{has property } (TT)/T.$$

For every $n \geq 2$, matrix self-similarity and the elementary diagonal reduction give

$$E_n(L_{\mathbb{F}_2}(1, 2)) = \mathrm{GL}_n(L_{\mathbb{F}_2}(1, 2)) \cong L_{\mathbb{F}_2}(1, 2)^\times.$$

In particular, every displayed elementary group has property $(TT)/T$.

Formal counterpart:

`PropertyTTPaper.binaryLeavittUnits_hasTTmodT`.

Formal counterpart:

`PropertyTTPaper.binaryLeavitt_elementaryEquivUnits`.

Formal counterpart:

`PropertyTTPaper.binaryLeavitt_elementaryGroup_hasTTmodT`.

The rank-two case is not a low-rank analogue of the universal-lattice argument. The analytic estimate is proved in rank four; the binary Leavitt self-similarity then identifies $E_2(L)$ with $E_4(L)$. This is why a rank-two elementary group can inherit property $(TT)/T$ here.

Khanh and Thanh prove that every complete binary leaf set C with $n \geq 2$ identifies

$$L_k(1, 2)^\times = \Theta_C(E_n(L_k(1, 2)))$$

[6, Corollary 4.4]. Their identification agrees with the rank collapse in Theorem C; the new conclusion here is property $(TT)/T$ for this unit group.

Relation to previous work. The argument combines several established theories. Mimura introduced property $(TT)/T$, developed the relative-to-global strategy for quasi-cocycles, and proved the noncommutative relative result discussed in Section 2 [7, 8]. The quantitative argument below gives an explicit root bound and also controls the invariant-vector summand. Ershov and Jaikin-Zapirain proved property (T) for elementary groups of rank at least three over finitely generated associative rings [4]. On the ring-theoretic side, Ara, Goodearl, and Pardo established the GE and unit-group K_1 structure of purely infinite simple rings [3], while Ara, Brustenga, and Cortiñas computed the stable algebraic K -theory of Leavitt path algebras [2].

The pivot in Section 3 belongs to the same Gaussian-elimination circle as the GE reduction of Ara–Goodearl–Pardo. The fixed-coordinate organization is the additional feature used here: the coordinate is chosen before the matrix, exactly one block factor occurs, and the remaining word has length at most $2n + 1$. Thus the full factorization has width $2n + 2$. The rigidity theorem uses this factorization to globalize relative quasi-cocycle bounds. Section 2 gives a quantitative proof of the root estimate, and Section 5 verifies the remaining structural hypotheses for the binary Leavitt algebra.

Organization. Section 2 develops the relative root estimate. Section 3 proves the diagonal pivot and the coordinate-block factorization. Section 4 combines the analytic and algebraic inputs to prove Theorem B, and Section 5 verifies its hypotheses for the binary Leavitt algebra. The final section discusses the natural extensions suggested by the proof.

2. QUASI-COCYCLES AND RELATIVE CONTROL

Let G be a group and let $\pi : G \rightarrow \mathcal{U}(\mathcal{H})$ be a unitary representation on a complex Hilbert space. A map $b : G \rightarrow \mathcal{H}$ is a *quasi-cocycle of defect at most $D \geq 0$* if

$$(2) \quad \|b(gh) - b(g) - \pi(g)b(h)\| \leq D \quad (g, h \in G).$$

An exact cocycle has $D = 0$; it is the translation part of an affine isometric action. A quasi-cocycle permits uniformly bounded failure of the affine action law.

Definition 2.1. A group G has property $(TT)/T$ if, for every unitary representation π with no nonzero invariant vector, every Hilbert-space quasi-cocycle into π is bounded on G . For a subgroup $H \leq G$, the pair (G, H) has relative property $(TT)/T$ if the same hypotheses imply boundedness on H . We write relative (TT) when the conclusion holds without excluding invariant vectors.

Mimura's qualitative route. Mimura proves that, for the free associative ring $R = \mathbb{Z}\langle X \rangle$, the pair

$$E_m(\mathbb{Z}\langle X \rangle) \geq E_{m-1}(\mathbb{Z}\langle X \rangle), \quad m \geq 3,$$

has relative property $(TT)/T$, with the smaller group in the upper-left corner [8, p. 154, Proposition 9.2.8]. Relative $(TT)/T$ descends through a surjective homomorphism: pull a representation and quasi-cocycle back along the quotient map and then apply the relative bound. Reduce coefficients modulo 2, take $m = 4$, and conjugate the standard rank-three block so that it contains X_{03} . This gives a short qualitative proof of relative $(TT)/T$ for $(E_4(\mathbb{F}_2\langle X \rangle), X_{03})$. We now prove the stronger relative (TT) statement, with an explicit bound.

We first isolate the quantitative rank-three estimate used in the direct proof. For distinct i, j, k , write $H_{ijk} = \langle X_{ik}, X_{jk} \rangle \leq E_3(A)$, and identify the elements of X with the corresponding free generators of A .

The proof uses the word-length filtration $A_{\leq d} \subseteq A$. Put $V_d = A_{\leq d} \oplus A_{\leq d}$, viewed as the finite elementary abelian subgroup

$$(a, c) \longmapsto x_{ik}(a)x_{jk}(c) \quad \text{of } H_{ijk}.$$

For an orthogonal representation σ on a real Hilbert space and $z \in \mathcal{H}$, let $z_{d,\chi}$ be the Fourier component of z belonging to $\chi \in \widehat{V}_d$. The components are orthogonal and their squared norms sum to $\|z\|^2$. Write a character as (χ_1, χ_2) , and let $\nu(\chi_r)$ be the least word degree on which χ_r is nontrivial, with value $d + 1$ for the trivial character. The nontrivial pairs split into

$$\begin{array}{ll} A & 0 < \nu(\chi_2) < \nu(\chi_1) \\ B & 0 < \nu(\chi_1) = \nu(\chi_2) \\ C & 0 < \nu(\chi_1) < \nu(\chi_2) \\ D & \nu(\chi_1) = 0 \text{ or } \nu(\chi_2) = 0. \end{array}$$

Let $m_d^A, m_d^B, m_d^C, m_d^D$ be the corresponding Fourier masses. Let $\partial_{1,d}$ and $\partial_{2,d}$ be the masses for which the indicated coordinate character is trivial on $A_{\leq d-1}$ and becomes nontrivial on $A_{\leq d}$. Equivalently, one may index Fourier components by Boolean sign assignments on a basis of V_d . A nonadditive assignment indexes the zero component, so only genuine characters contribute to the transport identities below.

Lemma 2.2 (Finite-plane energy estimate). *Let $\sigma : E_3(A) \rightarrow \mathcal{O}(\mathcal{H})$ be an orthogonal representation on a complete real Hilbert space. If P is the orthogonal projection onto $\mathcal{H}^{H_{ijk}}$ and $w = z - Pz$, then*

$$\begin{aligned} \|w\|^2 &\leq d_{ik,1}^2 + d_{jk,1}^2 + \frac{3}{2} \sum_{s \in X} 2 \|z\| (d_{ij,s} + d_{ji,s}) \\ (3) \quad &+ 2 \|z\| (d_{ij,1} + d_{ji,1}), \end{aligned}$$

where $d_{ab,t} = \|\sigma(x_{ab}(t))z - z\|$.

Formal counterpart:

[PropertyTTPaper.freeCharacteristicTwo_planeEnergyEstimate.](#)

Proof. Set

$$U_{ab} = 2 \|z\| d_{ab,1}, \quad G_{ab} = \sum_{s \in X} 2 \|z\| d_{ab,s}.$$

Conjugation by $x_{ji}(1)$ shears region A into region B , and conjugation by $x_{ij}(1)$ does the same from C to B . More explicitly, on the plane coordinates these conjugations act by

$$(a, c) \mapsto (a, c + a), \quad (a, c) \mapsto (a + c, c),$$

with the choice of formula determined by the adjacent root. After precomposition, the corresponding pair of characters is transformed by the same elementary shear. Fourier covariance, orthogonality, and

$$\left| \sum_{\chi \in \Omega} \|z_{d,\chi}\|^2 - \sum_{\chi \in \Omega} \|z'_{d,\chi}\|^2 \right| \leq (\|z\| + \|z'\|) \|z - z'\|$$

for $z' = \sigma(g)z$ give

$$(4) \quad m_d^A \leq m_d^B + U_{ji}, \quad m_d^C \leq m_d^B + U_{ij}.$$

On region D , one coordinate character detects the unit coefficient. The nontrivial character of $\mathbb{F}_2$ has character gap 4, so the negative-character projection formula for an involution gives

$$(5) \quad m_d^D \leq \frac{1}{4}(d_{ik,1}^2 + d_{jk,1}^2).$$

We next record the cross-stage inequalities controlling the equal-valuation region. Write $\text{res}_d : \widehat{V}_{d+1} \rightarrow \widehat{V}_d$ for restriction. For each $\chi \in \widehat{V}_d$, its Fourier component is the orthogonal sum of the components indexed by $\text{res}_d^{-1}(\chi)$. Consequently its squared norm is the sum of their squared norms. A character whose two valuations are not newly exposed in degree $d + 1$ remains in the

same valuation region after restriction. Therefore, if m_d^S denotes the mass in a union S of valuation regions,

$$(6) \quad m_d^S \leq m_{d+1}^S + \partial_{1,d+1} + \partial_{2,d+1}.$$

The two generator shears give the sharper descents

$$(7) \quad m_{d+2}^{A \cup B} \leq m_{d+1}^{C \cup D} + G_{ji} + \partial_{2,d+2},$$

$$(8) \quad m_{d+2}^{C \cup B} \leq m_{d+1}^{A \cup D} + G_{ij} + \partial_{1,d+2}.$$

We prove the first; the second follows after interchanging the two coordinates. Fix an enumeration of X . For $s \in X$, the relevant shear is the homomorphism

$$\psi_s : V_{d+1} \longrightarrow V_{d+2}, \quad (a, c) \longmapsto (a, c + sa).$$

It is induced by conjugation with $x_{ji}(s)$. Notice that its domain and codomain are different: multiplication by s raises degree, so no invariance of a finite truncation is being asserted. Pullback along ψ_s sends a character pair $(\chi_1, \chi_2) \in \hat{V}_{d+2}$ to

$$\psi_s^*(\chi_1, \chi_2) = (\chi_1 \partial_s \chi_2, \chi_2)$$

where $(\partial_s \chi_2)(a) = \chi_2(sa)$. The leading-degree $\nu(\partial_s \chi_2) = \nu(\chi_2) - 1$.

Now partition the characters in $A \cup B$ at stage $d+2$ by the least generator s for which the leading part of $\partial_s \chi_2$ is nontrivial. First remove the characters whose second coordinate is detected for the first time in degree $d+2$; their mass is the boundary term $\partial_{2,d+2}$. For every remaining character, the leading part of χ_2 survives restriction to $A_{\leq d+1}$. Its pullback therefore lies in $C \cup D$, and the least-generator rule can be read back from that surviving leading part. Hence the receiving families for distinct s are disjoint. Fourier covariance identifies the spectral projection for a source fiber with the projection for its pullback, up to replacing z by $\sigma(x_{ji}(s))z$. The variation inequality then charges the fiber to a single copy of $m_{d+1}^{C \cup D}$, with error $2\|z\| d_{ji,s}$. Summing the disjoint fibers over $s \in X$ proves (7). The forward shear $(a, c) \mapsto (a + sc, c)$, whose character pullback is $(\chi_1, \chi_2 \partial_s \chi_1)$, proves (8).

Add (7) and (8), use (6) to move their coarse masses to stage $d+2$, and then use (4) and (5). Collecting the four region coefficients gives

$$(9) \quad m_{d+2}^A + m_{d+2}^B + m_{d+2}^C + m_{d+2}^D \leq d_{ik,1}^2 + d_{jk,1}^2 + \frac{3}{2}(G_{ji} + G_{ij}) + U_{ji} + U_{ij} + \frac{9}{2}(\partial_{1,d+2} + \partial_{2,d+2}).$$

This is the finite-stage estimate whose limit is (3); it also accounts for the coefficient $3/2$ there.

The boundary limit follows from an exact telescoping identity. If $T_{r,d}$ is the mass on which the r -th coordinate character is trivial on $A_{\leq d}$, orthogonal refinement gives

$$T_{r,d} = \partial_{r,d+1} + T_{r,d+1}.$$

The nonnegative sequence $T_{r,d}$ decreases, so $\partial_{r,d+1} = T_{r,d} - T_{r,d+1} \rightarrow 0$. Finally, the subgroups V_d increase to H_{ijk} . The orthogonal projections onto their moving subspaces converge strongly to the moving projection for H_{ijk} ;

equivalently, the decreasing fixed spaces have intersection $\mathcal{H}^{H_{ijk}}$. Hence the left side of (9) converges to $\|z - Pz\|^2$. Passing to the limit proves (3). $\square$

Lemma 2.3 (Localized plane estimate). *Let $\sigma : E_3(A) \rightarrow \mathcal{U}(\mathcal{H})$ be a unitary representation and let $z \in \mathcal{H}$. Suppose $B \geq 0$ and the displacement of z is at most B under*

$$x_{ik}(1), x_{jk}(1), x_{ij}(1), x_{ji}(1), x_{ij}(s), x_{ji}(s) \quad (s \in X).$$

Then, for every $g \in H_{ijk}$,

$$\|\sigma(g)z - z\| \leq 2(6|X| + 6)(B + 1).$$

Formal counterpart:

[PropertyTTPaper.freeCharacteristicTwo_localizedPlaneEstimate.](#)

Proof. Regard the complex Hilbert space as real and the unitary operators as orthogonal operators; norms, fixed spaces, and displacements are unchanged. Put $\delta = B + 1$, let P be the orthogonal projection onto $\mathcal{H}^{H_{ijk}}$, and set $w = z - Pz$. The adjacent root groups X_{ij} and X_{ji} normalize H_{ijk} , while $X_{ik}, X_{jk} \leq H_{ijk}$. Orthogonal projection therefore passes each of the displayed displacement bounds from z to w , with the strict bound δ . Apply Lemma 2.2 to w . Since $Pw = 0$, it gives

$$\|w\|^2 \leq 2\delta^2 + (6|X| + 4)\|w\|\delta.$$

If $\|w\| \leq \delta$, the desired bound on w is immediate. Otherwise $\delta \leq \|w\|$, and the last inequality gives $\|w\| \leq (6|X| + 6)\delta$. Finally, every $g \in H_{ijk}$ fixes Pz , so $\|\sigma(g)z - z\| \leq 2\|w\|$. $\square$

Lemma 2.4 (Row-plane symmetry). *Let $\bar{\cdot} : A \rightarrow A$ reverse every word and fix the free generators. The inverse-transpose map*

$$\Theta(g) = \bar{g}^{-t}$$

is an automorphism of $E_3(A)$, and

$$\Theta(x_{ij}(a)) = x_{ji}(-\bar{a}) \quad (a \in A).$$

In particular, $\Theta(x_{ij}(t)) = x_{ji}(t)$ for $t \in \{1\} \cup X$. Consequently Lemma 2.3 applies with the two root directions reversed.

Formal counterpart:

[PropertyTTPaper.freeCharacteristicTwo_inverseTranspose_root.](#)

Formal counterpart:

[PropertyTTPaper.freeCharacteristicTwo_rowEmbedding_controlRoot.](#)

Formal counterpart:

[PropertyTTPaper.freeCharacteristicTwo_rowEmbedding_elementaryRoot.](#)

Proof. Word reversal followed by transpose identifies the matrix ring with its opposite ring. On units, inversion identifies the opposite group with the original group, so Θ is an automorphism. Since $E_{ij}^2 = 0$, $x_{ij}(a)^{-1} = x_{ij}(-a)$. In characteristic two the sign vanishes, while word reversal fixes 1 and every $s \in X$, which gives the displayed formula. $\square$

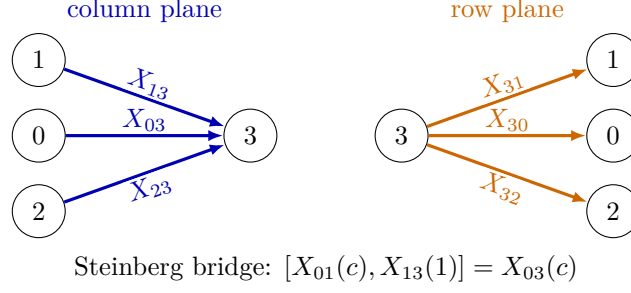


FIGURE 1. The rank-four root geometry. The two rank-three plane estimates control the blue column roots and, through Lemma 2.4, the orange row roots. The displayed Steinberg commutator recovers the target root X_{03} .

Theorem 2.5 (Root quasi-cocycle estimate). *Let X be finite, put $A = \mathbb{F}_2\langle X \rangle$, and let*

$$G = E_4(A), \quad X_{03} = \{I + aE_{03} : a \in A\}.$$

For every unitary representation π , every $D \geq 0$, and every quasi-cocycle $b : G \rightarrow \mathcal{H}$ of defect at most D , the set $b(X_{03})$ is bounded.

Formal counterpart:

[PropertyTTPaper.freeCharacteristicTwo_root_hasRelativeTT.](#)

Proof. Let N be the last-column subgroup generated by X_{03}, X_{13}, X_{23} . Decompose

$$\mathcal{H} = \mathcal{H}^N \oplus (\mathcal{H}^N)^\perp, \quad b(r) = p + w$$

for $r = I + aE_{03}$. Since the coefficient ring has characteristic two, $r^2 = 1$. First, $\|b(1)\| \leq D$, by applying (2) to $1 \cdot 1$. Apply the same inequality to r^2 and project orthogonally to $\mathcal{H}^N$. Since $r \in N$, it acts as the identity on this subspace. If q denotes the projection of $b(1)$, contractivity of the projection gives

$$\|q - p - p\| \leq D, \quad \|q\| \leq D.$$

Thus $\|2p\| \leq 2D$, and hence

$$(10) \quad \|p\| \leq D.$$

For the moving component, let $Q_X \subseteq E_3(A)$ consist of all elementary roots with coefficient 1 or one of the free generators. Embed it on the coordinates $\{1, 2, 3\}$, and also on the coordinates $\{0, 1, 2\}$ using the transpose-word-reversal symmetry of Lemma 2.4; the geometry is shown in Figure 1. Put

$$M = \sum_{s \in Q_X} (\|b(\iota_{123}(s))\| + \|b(\iota_{012}(s))\|), \quad B = 2M + 2D,$$

$$K = 2(6|X| + 6)(B + 1).$$

Only the following members of these two finite embedded sets are used:

$$\begin{aligned} & x_{13}(1), x_{23}(1), x_{12}(t), x_{21}(t), \\ & x_{01}(1), x_{02}(1), x_{21}(t), x_{12}(t), \end{aligned} \quad t \in \{1\} \cup X.$$

Here the first row comes from the 123-embedding and the second from the 012-embedding. Every displayed element commutes with $r = x_{03}(a)$ and normalizes N . Comparing the quasi-cocycle equations for sr and rs , then projecting to the moving subspace, gives for each displayed control element

$$\|\pi(s)w - w\| \leq \|\pi(s)b(r) - b(r)\| \leq 2\|b(s)\| + 2D \leq B.$$

Lemma 2.3 propagates these finitely many inequalities to the complete root planes in the two embeddings. In particular, for every $c \in A$,

$$\|\pi(x_{13}(c))w - w\|, \quad \|\pi(x_{23}(c))w - w\|, \quad \|\pi(x_{01}(c))w - w\| \leq K.$$

The Steinberg relation

$$[I + cE_{01}, I + E_{13}] = I + cE_{03}$$

and the unitary commutator estimate give displacement at most $4K$ under $x_{03}(c)$. Every element of N is a product of one element from each of X_{03}, X_{13}, X_{23} , so its displacement of w is at most $6K$. Since $w \perp \mathcal{H}^N$, this already bounds w . Indeed, the closed convex hull of the N -orbit of w has a unique vector of minimal norm; that vector is N -fixed. The convex hull stays inside $(\mathcal{H}^N)^\perp$, so its minimal vector is zero. On the other hand, the whole orbit, and hence its closed convex hull, lies in the closed ball of radius $6K$ about w . Consequently $\|w\| \leq 6K$. Together with (10), this gives

$$\|b(I + aE_{03})\| \leq D + 12(6|X| + 6)(2M + 2D + 1),$$

uniformly in $a \in A$. $\square$

Remark 2.6 (A qualitative proof from Mimura's theorem). For qualitative boundedness, Mimura's relative theorem gives a shorter proof of Theorem 2.5. Split $\mathcal{H} = \mathcal{H}^G \oplus (\mathcal{H}^G)^\perp$. Mimura's relative theorem bounds the second projection on X_{03} . On the invariant summand the action is trivial, while $r^2 = 1$ for $r \in X_{03}$; the quasi-cocycle identities for 1 and r^2 then give $\|b_0(r)\| \leq D$. The direct proof above complements this argument by recording an explicit bound and the quantitative rank-three propagation behind it.

3. COORDINATE-BLOCK FACTORIZATION

We now prove the main algebraic input. Write

$$e_{ij}(a) = I + aE_{ij} \quad (i \neq j)$$

for an elementary transvection.

Lemma 3.1 (Three-move diagonal pivot). *Let R satisfy (1), let $A \in \mathrm{GL}_n(R)$, and fix distinct coordinates i, j . At most three elementary row and column operations in total produce a matrix B satisfying*

$$B_{jj} = 1.$$

More precisely, one left factor and two right factors suffice.

Formal counterpart:

[PropertyTTPaper.threeMove_diagonalPivot.](#)

Proof. We give the calculation, since it is the source of the uniform constant. Choose two distinct indices i, j . First arrange that the entry $a = A_{ii}$ is nonzero. If it is zero, invertibility of A gives k with $A_{ik} \neq 0$; right multiplication by $e_{ki}(1)$ replaces A_{ii} by A_{ik} . This costs at most one preliminary column operation.

Write $b = A_{ij}$, and choose $p, q \in R$ with $paq = 1$. Put

$$\rho = q(1 - pb), \quad F_1 = e_{ij}(\rho), \quad A_1 = AF_1, \quad b_1 = (A_1)_{ij}.$$

Then

$$b_1 = a\rho + b, \quad pb_1 = paq(1 - pb) + pb = 1.$$

Set

$$d_1 = (A_1)_{jj}, \quad w = (1 - d_1)p, \quad E_1 = e_{ji}(w), \quad A_2 = E_1A_1.$$

The (j, j) -entry is now

$$(A_2)_{jj} = d_1 + wb_1 = d_1 + (1 - d_1)pb_1 = 1.$$

Thus $B = A_2 = E_1AF_1$ has $B_{jj} = 1$. This uses one row and one column transvection. Including the possible preliminary column addition gives one left factor and at most two right factors. The calculation applies to arbitrary rings satisfying (1). $\square$

Proof of Theorem A. Choose $i \neq j$. Lemma 3.1 gives elementary products E, F , represented by lists ℓ, r , such that

$$B = EAF, \quad B_{jj} = 1,$$

with

$$|\ell| \leq 1, \quad |r| \leq 2.$$

For every $k \neq j$, right multiply by

$$e_{jk}(-B_{jk}).$$

These transvections have a common source column and distinct target columns, so each previously cleared entry stays zero. If T is their product and $C = BT$, then

$$C_{jc} = \begin{cases} 1, & c = j, \\ 0, & c \neq j. \end{cases}$$

The list t of right-clearing factors has length exactly $n - 1$.

For every $k \neq j$, left multiply by

$$e_{kj}(-C_{kj}).$$

Again the operations have distinct target rows and preserve the previously cleared entries. If S is their product and $D = SC$, then

$$D_{rc} = \begin{cases} 1, & r = c = j, \\ 0, & r = j, c \neq j, \\ 0, & r \neq j, c = j, \\ C_{rc}, & r, c \neq j. \end{cases}$$

Hence $D \in \mathrm{GL}_{n-1}^{(j)}(R)$. The list s of left-clearing factors also has length exactly $n - 1$.

Solving $D = SEAF$ for A gives

$$A = E^{-1}S^{-1}DT^{-1}F^{-1}.$$

At the word level, inverse products are represented by reversing the factor lists and inverting every transvection. The inverse of $e_{ab}(z)$ is $e_{ab}(-z)$, so all these factors remain elementary. The resulting word has the exact form

$$\ell_{\mathrm{rev}}^{-1} s_{\mathrm{rev}}^{-1} [D] t_{\mathrm{rev}}^{-1} r_{\mathrm{rev}}^{-1},$$

and therefore length at most

$$1 + (n - 1) + 1 + (n - 1) + 2 = 2n + 2.$$

□

Remark 3.2. The fixed coordinate j is chosen before A . Allowing the coordinate block to depend on A gives a different, weaker width statement; the fixed-coordinate form is what feeds the uniform quasi-cocycle globalization below.

4. FROM RELATIVE CONTROL TO GLOBAL RIGIDITY

Let R be a finite-type $\mathbb{F}_2$ -algebra. Choose a finite set X and a surjection

$$\mathbb{F}_2\langle X \rangle \twoheadrightarrow R.$$

Elementary matrices are functorial in the coefficient ring, so the root estimate of Theorem 2.5 descends to R . Coordinate permutations then give the same estimate for every root subgroup in rank four.

Two larger pieces are used in the assembly: a last-column subgroup and a last-row subgroup. The quasi-cocycle is bounded on both. The stabilized rank-three coordinate group normalizes these pieces. Kazhdan control for the rank-three group converts boundedness on the normalized root system into boundedness on the coordinate group, provided the ambient representation has no invariant vectors. This is the only place where the quotient by the trivial representation enters.

We record the relative-to-global step explicitly. For a subset $S \subseteq G$, say that quasi-cocycles are *bounded on S* if every quasi-cocycle into a unitary representation without invariant vectors has bounded restriction to S .

Lemma 4.1 (Kazhdan globalization on a normalized set). *Let G have property (T), let $S \subseteq G$ generate G , and let $H \leq G$. Suppose*

$$h^{-1}Sh \subseteq S \quad (h \in H)$$

and quasi-cocycles are bounded on S . Then (G, H) has relative property $(TT)/T$.

Formal counterpart:

[PropertyTTPaper.kazhdan_normalizedSet_globalization](#).

Proof. Fix a unitary representation π of G without nonzero invariant vectors and a quasi-cocycle b of defect at most D . Choose a Kazhdan pair (K, ε) , and choose once and for all an $S \cup S^{-1}$ -word for each $k \in K$. Let L be the sum of their lengths. If $\|b(s)\| \leq C$ for $s \in S$, comparison of the quasi-cocycle identities for sh and $h(h^{-1}sh)$ gives

$$\|\pi(s)b(h) - b(h)\| \leq 2C + 2D \quad (s \in S \cup S^{-1}, h \in H).$$

For an inverse letter, apply the comparison to its inverse. The identity

$$\|\pi(s^{-1})v - v\| = \|\pi(s)v - v\|$$

shows that the constant remains $2C + 2D$. The displacement estimate for a product of unitaries therefore gives

$$\|\pi(k)b(h) - b(h)\| \leq L(2C + 2D) \quad (k \in K).$$

Since π has no invariant vectors, the Kazhdan inequality supplies some $k \in K$ with $\varepsilon \|b(h)\| \leq \|\pi(k)b(h) - b(h)\|$. Thus $\|b(h)\| \leq L(2C + 2D)/\varepsilon$, uniformly in $h \in H$. $\square$

Lemma 4.2 (Leavitt rank equivalence). *If R contains a binary Leavitt family, then for all $m, n \geq 2$ there is a group isomorphism*

$$E_m(R) \cong E_n(R).$$

Formal counterpart:

[PropertyTTPaper.leavitt_elementaryRankEquivalence](#).

Formal counterpart:

[PropertyTTPaper.elementaryBlockFlattening](#).

Proof. Iterating the binary prefix code gives a ring isomorphism $M_q(R) \cong R$ for every $q \geq 2$. Functoriality in the coefficient ring and block flattening give

$$E_m(R) \cong E_m(M_n(R)) \cong E_{mn}(R) \cong E_n(M_m(R)) \cong E_n(R).$$

The middle identifications are bidirectional on elementary groups. A block transvection decomposes into commuting scalar transvections after flattening. Conversely, a scalar transvection whose indices lie in different outer blocks is already a block transvection. If both indices lie in one outer block, choose a second outer block and use the Steinberg identity

$$e_{(i,k),(i,l)}(a) = [e_{(i,k),(j,k)}(a), e_{(j,k),(i,l)}(1)].$$

Thus block flattening identifies the full elementary subgroups in both directions, and the displayed chain restricts to the claimed group isomorphism. $\square$

Lemma 4.3 (Elementary equals general linear). *Assume that R has a binary Leavitt family, satisfies (1), and satisfies $\text{diag}(u, 1) \in E_2(R)$ for every $u \in R^\times$. Then*

$$E_n(R) = GL_n(R) \quad (n \geq 2).$$

Formal counterpart:

[PropertyTTPaper.elementaryGroup_eq_generalLinear](#).

Proof. The single-sandwich elimination reduces an arbitrary element of $GL_2(R)$, by elementary row and column operations, to $\text{diag}(u, 1)$ for some $u \in R^\times$. The diagonal hypothesis therefore gives $GL_2(R) = E_2(R)$.

The binary Leavitt family gives a coefficient isomorphism $M_n(R) \cong R$ for every $n \geq 2$. Transporting the two ring-theoretic hypotheses across this isomorphism gives the same rank-two equality over $M_n(R)$. The bidirectional block flattening proved in Lemma 4.2 then gives

$$E_2(M_n(R)) = GL_2(M_n(R)) \implies E_{2n}(R) = GL_{2n}(R).$$

Equivalently, one may write the same flattening after interchanging the two finite index factors as

$$E_n(M_2(R)) = GL_n(M_2(R)).$$

Transport through $M_2(R) \cong R$ now gives $E_n(R) = GL_n(R)$. $\square$

Proposition 4.4 (Rank-four assembly). *Let X be finite and let $f : \mathbb{F}_2\langle X \rangle \twoheadrightarrow R$ be a surjective ring homomorphism. Suppose that R is nontrivial, contains a binary Leavitt family, satisfies (1), and satisfies $\text{diag}(u, 1) \in E_2(R)$ for every $u \in R^\times$. Then $E_4(R)$ has property $(TT)/T$.*

Formal counterpart:

[PropertyTTPaper.finiteFreeQuotient_rankFour_hasTTmodT](#).

Proof. Theorem 2.5 descends through f and gives relative $(TT)/T$ for $X_{03}(R)$. Coordinate permutations give it for the six roots

$$X_{03}, X_{13}, X_{23}, X_{30}, X_{31}, X_{32}.$$

The assembly follows the chain

$\begin{aligned} &\text{six root bounds} \implies \text{last row and last column are bounded} \\ &\implies \text{the rank-three core is bounded} \implies \text{coordinate blocks are bounded} \\ &\xRightarrow{\text{Theorem A}} \text{the whole rank-four group is bounded.} \end{aligned}$

Here are the details. Put

$$U^+ = X_{03}X_{13}X_{23}, \quad U^- = X_{30}X_{31}X_{32}.$$

The three roots in each product commute, so the quasi-cocycle identity makes b bounded on U^+ and on U^- . The set $S = U^+ \cup U^-$ generates $E_4(R)$: roots

meeting the last coordinate lie in S , and every remaining root is a Steinberg commutator of one last-column and one last-row root.

For the Kazhdan input, the finite set of elementary roots with coefficients in $\{1\} \cup X$ is a Kazhdan set for $E_3(\mathbb{F}_2\langle X \rangle)$: the rank-three root estimate gives a Kazhdan constant after division by $2(12|X| + 13)$. Descending this finite Kazhdan pair through f gives property (T) for $E_3(R)$. This is the finite-control form of the Ershov–Jaikin–Zapirain theorem used here [4]. Lemma 4.2 transports property (T) to $E_4(R)$. The stabilized rank-three subgroup $H = E_3(R)$ normalizes both U^+ and U^- , hence normalizes S . Lemma 4.1 now proves that $(E_4(R), H)$ has relative property $(TT)/T$. Notice that the no-invariant-vectors hypothesis is used precisely in the Kazhdan inequality inside that lemma.

By Lemma 4.3, the elementary groups in ranks three and four are the full general linear groups. Under these identifications the stabilized core is exactly the coordinate block $\mathrm{GL}_3^{(3)}(R)$. Finally, coordinate permutations carry this relative statement to $\mathrm{GL}_3^{(j)}(R)$ for every $j \in \{0, 1, 2, 3\}$.

Each elementary transvection lies in a coordinate block avoiding its two indices. Hence the union of one fixed coordinate block and all root subgroups is bounded for the quasi-cocycle. Theorem A writes every element of $\mathrm{GL}_4(R)$ as a product of at most 10 elements of that union. Lemma 4.5 below, or directly the corresponding bounded-product lemma, makes the quasi-cocycle bounded on $\mathrm{GL}_4(R)$. Transporting through $E_4(R) = \mathrm{GL}_4(R)$ proves the proposition. $\square$

The passage from a bounded generating subset to the whole group uses the following elementary estimate.

Lemma 4.5 (Products of controlled pieces). *Let b be a quasi-cocycle of defect at most D , and suppose $\|b(g_i)\| \leq C$ for a list $(g_1, \dots, g_m)$. Then*

$$\|b(g_1 \cdots g_m)\| \leq mC + (m+1)D.$$

Formal counterpart:

`PropertyTPaper.quasiCocycle_list_product_bound.`

Proof. Induct on the list. The empty-list case is $\|b(1)\| \leq D$. For a nonempty list, apply (2) to the head and the product of the tail, use that the representation is isometric, and invoke the inductive bound. The displayed estimate is a slightly loose uniform form that also covers the empty list. $\square$

Proof of Theorem B. Finite type supplies a finite set X and a surjection $\mathbb{F}_2\langle X \rangle \twoheadrightarrow R$, so Proposition 4.4 gives the result in rank four. For arbitrary $n \geq 2$, Lemma 4.2 supplies an explicit group isomorphism $E_n(R) \cong E_4(R)$. Property $(TT)/T$ is invariant under group isomorphism, so the rank-four conclusion transports to rank n . $\square$

5. THE BINARY LEAVITT SPECIALIZATION

Let L be the binary Leavitt algebra [5],

$$L = L_{\mathbb{F}_2}(1, 2) = \mathbb{F}_2\langle x_0, x_1, y_0, y_1 \rangle / (y_i x_j - \delta_{ij}, x_0 y_0 + x_1 y_1 - 1).$$

The defining generators themselves form a binary Leavitt family, and the presentation makes L a finite-type $\mathbb{F}_2$ -algebra.

For a binary word $\alpha = i_1 \cdots i_r$, write $x_\alpha = x_{i_1} \cdots x_{i_r}$ and $y_\alpha = y_{i_r} \cdots y_{i_1}$. Then $y_\alpha x_\beta$ is determined by prefix comparison, and at every depth r ,

$$(11) \quad \sum_{|\delta|=r} x_\delta y_\delta = 1.$$

Lemma 5.1 (Single-sandwich division for $L_k(1, 2)$). *For every field k and every nonzero $z \in L_k(1, 2)$, there are $a, b \in L_k(1, 2)$ such that $azb = 1$.*

Formal counterpart:

`BinaryLeavitt.exists_mul_mul_eq_one.`

Proof. Write z as a finite sum $\sum_i c_i x_{\alpha_i} y_{\beta_i}$, and let $M = \max_i |\beta_i|$. Equation (11) shows that $zx_\delta \neq 0$ for some word δ of length M ; otherwise $z = \sum_{|\delta|=M} zx_\delta y_\delta = 0$. Prefix cancellation expresses zx_δ as a finite linear combination of pure x -words. After collecting equal words, write

$$zx_\delta = \sum_{\gamma \in G} c_\gamma x_\gamma, \quad c_\gamma \neq 0,$$

and choose $\gamma_0 \in G$ of minimal length. Multiplication on the left by y_{γ_0} turns the γ_0 -term into c_{γ_0} ; every other surviving term is a proper x_ε , with $1 \leq |\varepsilon| \leq m$ for $m = \max_{\gamma \in G} |\gamma|$.

Let $w = 0^m 1$. For every such ε , the words w and εw are incomparable: they disagree in position m , while εw is longer than w . Hence $y_w x_\varepsilon x_w = 0$. It follows that

$$y_w y_{\gamma_0} z x_\delta x_w = c_{\gamma_0}.$$

Thus

$$(c_{\gamma_0}^{-1} y_w y_{\gamma_0}) z (x_\delta x_w) = 1.$$

□

This direct argument proves the single-sandwich form of pure infinite simplicity used below. Alternatively, the graph criterion of Abrams–Aranda Pino gives pure infinite simplicity, and Ara–Goodearl–Pardo’s characterization gives the same conclusion [1, Theorem 11] [3, Theorem 1.6]. It remains to establish the unstable diagonal condition.

Proposition 5.2 (Unstable diagonal reduction). *For every field k , write $L_k = L_k(1, 2)$. For every $u \in L_k^\times$,*

$$\text{diag}(u, 1) \in E_2(L_k).$$

Consequently $\text{GL}_2(L_k) = E_2(L_k)$.

Formal counterpart:

`KOnePaper.diagUnit_mem_elementary.`

Formal counterpart:

[KOnePaper.elementaryGroup_two_eq_top](#).

Proof. Put

$$\mathcal{W} = \{u \in L_k^\times : \text{diag}(u, 1) \in E_2(L_k)\}.$$

The elementary identities for diagonal stabilization show that $\mathcal{W}$ is a normal subgroup of $L_k^\times$.

Give L_k its standard $\mathbb{Z}$ -grading, in which a monomial $x_\alpha y_\beta$ has degree $|\alpha| - |\beta|$. The first reduction, by elementary code changes and square-zero tail removals, shows that every unit has a representative modulo $\mathcal{W}$ in the degree window $[-1, 1]$. Such a representative admits a *pencil form*: relative to finite complete binary prefix codes P, Q , every corner coefficient is

$$a_0 y_0 + a_1 y_1 + c + b_0 x_0 + b_1 x_1, \quad a_0, a_1, b_0, b_1, c \in k.$$

After refining P , its cardinality may be taken to be 2^r .

It remains to eliminate a pencil unit. Form the two coefficient matrices B_0, B_1 belonging to the positive-degree terms and stack them vertically. There are two cases.

If the stack has a nonzero kernel vector, an invertible scalar column operation makes one column simultaneously B_0 - and B_1 -free. The corresponding code-change unit lies in $\mathcal{W}$. Splitting that free code word into its two children replaces Q by a complete code with one additional word, leaves the row code fixed, and preserves the pencil form. Iteration is finite: while $|Q| < 2^{r+1}$, the integer $2^{r+1} - |Q|$ decreases strictly. When $|Q| \geq 2^{r+1}$, reshape to a row code of depth at most r and a column code of depth at least $r + 1$. Every resulting term has nonpositive degree, and the nonpositive-window elementary reduction places the unit in $\mathcal{W}$.

If instead the stack has a scalar left inverse, the inverse pencil has strictly negative degree after a complementary shallow/deep code reshape. The same nonpositive-window reduction puts the inverse, and hence the original unit, in $\mathcal{W}$. These two exits prove that every narrow-window unit belongs to $\mathcal{W}$. Combined with the first reduction, this gives $\mathcal{W} = L_k^\times$, which is the displayed diagonal assertion.

Finally, the single-sandwich elimination for a 2×2 invertible matrix produces elementary matrices E, F and a unit u with $EAF = \text{diag}(u, 1)$. Since all three factors on the right are elementary, A is elementary. Thus $\text{GL}_2(L_k) = E_2(L_k)$. $\square$

Remark 5.3 (Alternative structural proof). The conclusion also follows structurally from the Leavitt-path-algebra exact sequence of Ara–Brustenga–Cortiñas, which gives $K_1(L_k) = 0$, together with the GE theorem of Menal–Moncasi and Ara–Goodearl–Pardo’s identification of K_1 with the abelianization of the unit group for purely infinite simple rings [2, 9, 3].

Proof of Theorem C. The binary Leavitt algebra is a nontrivial finite-type $\mathbb{F}_2$ -algebra, its canonical generators form a binary Leavitt family, Lemma 5.1 supplies (1), and Proposition 5.2 supplies the elementary diagonal condition.

Theorem B gives property $(TT)/T$ for $E_n(L)$ in every rank $n \geq 2$, while Lemma 4.3 identifies it with $GL_n(L)$.

For each $n \geq 1$, the left-comb prefix code

$$0, 10, 110, \dots, 1^{n-2}0, 1^{n-1}$$

is a complete binary prefix code with n words. Its Leavitt matrix units give a ring isomorphism $M_n(L) \cong L$, hence a group isomorphism $GL_n(L) \cong L^\times$. For $n \geq 2$ this yields $E_n(L) \cong L^\times$. Transporting the rank-two conclusion across this isomorphism gives property $(TT)/T$ for $L^\times$. $\square$

6. FURTHER DIRECTIONS

The characteristic-two hypothesis enters most visibly in (10), where every root element has order two. Over a field of characteristic p , the corresponding root has order p , and iterating the quasi-cocycle identity gives

$$b(1) \approx b(r) + \pi(r)b(r) + \dots + \pi(r)^{p-1}b(r).$$

On the root-fixed space this is p times the fixed projection of $b(r)$, with error at most $(p-1)D$. Thus the fixed-space argument has a plausible finite-characteristic analogue. Extending the theorem requires uniform replacements for the characteristic-two rank-three control and the sign-free coordinate symmetries.

The algebraic factorization theorem suggests a separate question. For which natural classes of simple or purely infinite rings does the single-sandwich condition hold, and when can the coordinate block in Theorem A itself be controlled by elementary subgroups? A satisfactory answer would extend the present synthesis from the Leavitt setting to a broader rigidity criterion. The binary-family hypothesis is restrictive: it gives an isomorphism of right modules $R \cong R^2$, hence $[R] = 2[R]$ in $K_0(R)$. For $L_k(1, d)$, the class of the unit generates $K_0 \cong \mathbb{Z}/(d-1)$ [2], so no binary Leavitt family can exist when $d > 2$. Theorem B therefore does not extend to the higher-arity Leavitt algebras without a different self-similarity input.

APPENDIX A. GUIDE TO THE FORMAL COMPANION

The accompanying Lean formalization is available in the [project repository](#). Except where indicated, declarations are in the `PropertyTTPaper` namespace. Each numbered result carries a visible link to its public declaration. The table below lists the main implementation declarations in the order used by the mathematical argument. It is intended as a reading path for users who want to inspect a calculation or reuse a proof component.

Argument	Principal Lean implementation declarations, in proof order
Finite-plane filtration and Lemma 2.2	<code>FreeRootCharacterValuation.</code> <code>sum_norm_planeRegionUnion_sq_le_succ_add_topBoundaries;</code> <code>sum_norm_planeABRegion_sq_le_target_add_error_add_boundary;</code> <code>sum_norm_planeCBRegion_sq_le_target_add_error_add_boundary;</code> <code>sum_planeRegionMass_nonzero_le_explicit_errors;</code> <code>tendsto_planeFirstTopBoundaryMass_succ_zero;</code> <code>tendsto_planeSecondTopBoundaryMass_succ_zero;</code> <code>tendsto_norm_planeMovingPart_sq</code>
Lemma 2.3	<code>norm_comp_columnPlane_sub_le_of_root_bounds</code>
Row-plane transport and Theorem 2.5	<code>FreeElementaryTranspose.elementaryTransposeEquiv_elementaryRoot;</code> <code>RankFourRootGeometry.embedRow012_elementaryRoot;</code> <code>RankFourRelativeTT.norm_root03_movingProjection_le;</code> <code>RankFourRelativeTT.norm_root03_fixedProjection_le;</code> <code>RankFourRelativeTT.hasRelativeTT_X03</code>
Theorem A	<code>exists_three_move_diagonal_pivot;</code> <code>rightClearFactors_clear_row;</code> <code>leftClearFactors_make_coordinateBlock;</code> <code>boundedProduct_coordinateBlockOrRoot</code>
Rank transport and rank-four assembly	<code>FreeElementaryPropertyT.freeElementary_hasKazhdanPropertyT;</code> <code>LeavittFamily.rankSuccEquiv_elementaryBlockEquiv;</code> <code>FiniteTypeLeavittTT.elementaryGroup_eq_top;</code> <code>RankFourGlobalization.hasRelativeTTmodT_coreRange_of_roots;</code> <code>FiniteTypeLeavittTT.rankFour_hasTTmodT;</code> <code>FiniteTypeLeavittTT.elementaryGroup_hasTTmodT</code>
Proposition 5.2	<code>BinaryLeavitt.exists_narrow_representative;</code> <code>BinaryLeavitt.exists_pencil_form;</code> <code>BinaryLeavitt.pencil_unit_mem_pow;</code> <code>BinaryLeavitt.narrowReduction_holds;</code> <code>BinaryLeavitt.stableUnits_eq_top_holds;</code> <code>BinaryLeavitt.glTwo_eq_elementary_holds</code>
Lemma 5.1	<code>LeavittFamily.eq_sum_mul_wordS_mul_wordT;</code> <code>BinaryLeavitt.killWord_not_prefix;</code> <code>BinaryLeavitt.exists_mul_mul_eq_one</code>

Trust surface and definitions. The executable audit `scripts/Audit.lean` traverses the kernel dependency closure of every declaration in the `GroupApproximation` namespace. It rejects `sorryAx`, compiler-trust axioms, and project axioms; the current audit succeeds with only `propext`, `Classical.choice`, and `Quot.sound`, the three classical axioms used by `Mathlib`. There are no `sorry` terms in the development.

The definitions used by the linked statements are in `Basic.lean` in the `PropertyTT` directory. They quantify over complete complex Hilbert spaces and unitary representations; `IsQuasiCocycle` includes $D \geq 0$ and (2); `HasTTmodT` excludes nonzero invariant vectors exactly as in Definition 2.1. The rank-three Kazhdan input is proved from an explicit finite Kazhdan pair; its implementation entry point appears in the rank-transport

row of the table. The single-sandwich and unstable diagonal inputs for $L_k(1, 2)$ are theorems, not hypotheses, in `LeavittSimplicity.lean` and `RefineLoopDischarge.lean`.

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